

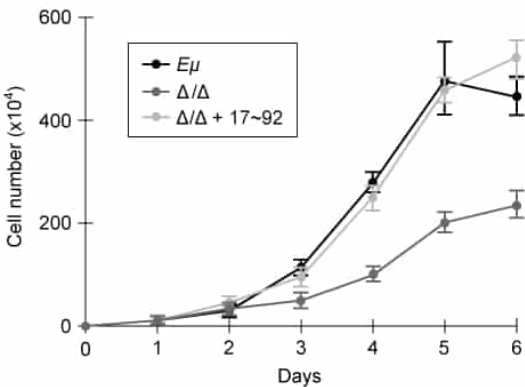
Overexpression of the *miR-17~92* gene cluster occurs in many human cancers involving *c-Myc*, a gene coding for a transcription factor that regulates expression of proteins such as p21, an inhibitor of cell cycle progression. *miR-17~92* encodes six distinct microRNAs (miRNAs) that can be divided into four families based on sequence identity at positions 2-7 (Table 1).

**Table 1** Sequence Identities of miRNA Families Encoded by the *miR-17~92* Gene Cluster

| miR family | Sequence identity | Encoded miRNA                   |
|------------|-------------------|---------------------------------|
| miR-17     | -AAAGUG-          | <i>miR-17</i> , <i>miR-20a</i>  |
| miR-18     | -AAGGUG-          | <i>miR-18a</i>                  |
| miR-19     | -GUGCAA-          | <i>miR-19a</i> , <i>miR-19b</i> |
| miR-92     | -AUUGCA-          | <i>miR-92</i>                   |

Experiment 1

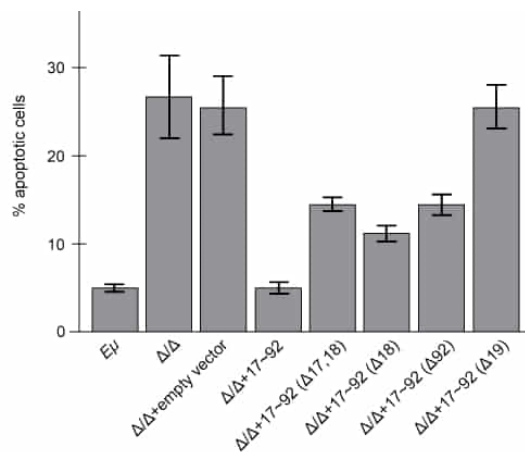
Two independent, diffuse B-cell lymphoma cell lines were used to study the functional relationship between *miR-17~92* transcripts and *c-Myc* in cancer pathogenesis. The first cell line was derived from *Eμ-Myc* mice (*Eμ*), which overexpress a *c-Myc* transgene controlled by a B-cell–specific enhancer that induces early development of B-cell lymphoma. The second cell line was derived from *Eμ-Myc* mice carrying two *miR-17~92* knockout alleles ( $\Delta/\Delta$ ). As shown in Figure 1, researchers measured the proliferation of both cell lines, as well as that of  $\Delta/\Delta$  cells infected with a retrovirus expressing the entire *miR-17~92* cluster ( $\Delta/\Delta + 17\sim92$ ).



**Figure 1** Growth curves of *Eμ* cells,  $\Delta/\Delta$  cells, and  $\Delta/\Delta + 17\sim92$  cells (error bars = standard deviation)

Experiment 2

To assess the relative contribution of each miRNA to the oncogenic ability of the *miR-17~92* cluster,  $\Delta/\Delta$  cell lines were transduced with an empty vector or with one of several *miR-17~92* mutant allele constructs that were engineered to lack expression of miRNA from at least one of the four families in the table. In each of the constructs, the family that was knocked out of the vector is indicated by the  $\Delta$  symbol. After verifying that the transduced cells correctly expressed the desired miRNA, the fraction of apoptotic cells in each line was measured (Figure 2).



**Figure 2** Percentage of apoptotic cells in *Eμ* cells and  $\Delta/\Delta$  cells transduced with the indicated *miR-17~92* construct (error bars = standard deviation)

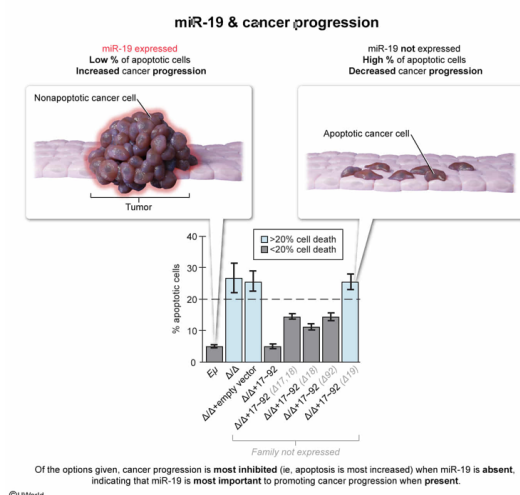
Based on Figure 2, which family from the *miR-17~92* gene cluster is expected to contribute the most to cancer progression?

- ☐ A. miR-17 [12%]
- ☐ B. miR-18 [34%]
- ☒ C. miR-19 [43%]
- ☐ D. miR-92 [9%]

Correct

43% answered correctly

### Explanation:



**Apoptosis**, or the programmed death of cells, limits the rate of cell proliferation. However, cancerous tumors develop as cells grow/divide **uncontrollably** and fail to follow predetermined apoptotic pathways. As a result, the **fraction of apoptotic cells** within a group of **cancer cells** is **markedly decreased (inhibited)**.

In *Experiment 2*, the researchers wanted to identify which miRNA family from the *miR-17~92* cluster contributed the most to apoptosis inhibition (ie, contributed most to cancer progression). First, a series of *miR-17~92* mutant allele constructs engineered to lack expression of specific miRNA families (indicated by  $\Delta$ ) were transduced into *Eμ-Myc* cells with the entire *miR-17~92*

cluster knocked out ( $\Delta/\Delta$ ). The fraction of apoptotic cells was then measured in these cell lines.

As shown in Figure 2, of all the cell lines transduced with nonempty vectors, the cell line expressing *all* miR-17~92 families *except* miR-19 (ie,  $\Delta/\Delta + 17\sim 92 [\Delta 19]$ ) had the *highest* percentage of apoptotic cells. This result signifies that **removal of only miR-19** from the *miR-17~92* cluster **promotes apoptosis** (ie, *inhibits* cancer progression) the *most*. Accordingly, when the miR-19 family is *present*, apoptosis is *suppressed* to the *greatest* extent (ie, cancer progression is *increased* the *most*). Therefore, of the options given, **miR-19** is likely **most essential to cancer progression** (ie, apoptosis inhibition).

**(Choice A)** To *precisely* conclude whether miR-17 has or lacks oncogenic properties, the percentage of apoptotic cells would have to be measured in *E $\mu$ -Myc* cell lines lacking miR-17 only ( $\Delta/\Delta + \Delta 17$ ), *not* in cells lacking both miR-17 and miR-18 (shown in Figure 2).

**(Choices B and D)** The cell line that lacks only miR-18 and the cell line that lacks only miR-92 still show a *low* percentage of apoptotic cells (characteristic of cancer progression). This finding means that *neither* miR-18 *nor* miR-92 contribute as much as miR-19 to promoting decreased apoptosis in cancer progression.

#### **Educational objective:**

Apoptosis is the programmed and controlled death of aged, unnecessary, or damaged cells. Cancerous cells exhibit rapid and uncontrolled proliferation, which is caused by cell growth that outpaces normal apoptosis or by dysfunction of the apoptotic process itself.

Time spent: 0 sec

Subject:  
Biology

QID: 400433

System:  
Molecular Biology